

The Transform Layer

Winding as Resampling, and Native Structure as Spectral Support

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Every numerical claim below was produced by code executed in the session that generated it. Residuals are quoted from live output. Section 11 is the falsification ledger, including two claims originated and killed inside this work and one correction to the predecessor paper's own methodological summary. Nothing is carried forward from an earlier draft without re-measurement.

Abstract

The Partner Predicate establishes that winding is transport rather than generation, and that lift heights multiply exactly. This paper supplies the mechanism, and shows the mechanism is stronger than the result it explains.

Under the realisation map $c_r = \lambda \hat{w}_r / (N \theta^r)$, subdivision by q is exactly zero insertion in the coefficient domain and exact periodic replication in the relation domain: $W_k = (1/q) w_{(k \bmod n)}$. The identity holds because $\theta_{(nq)}^q = \theta_n$ — the radical rescaling compensates the index rescaling, so the transform commutes with the winding map. Verified at (16,24) to 7.6×10^{-140} , inverted to 6.3×10^{-140} , forward-predicted on the previously unmeasured (16,16), and, decisively, verified on the hexagonal lineage at (64,18) to 6.1×10^{-199} . The two-factor gate of the predecessor paper is an octal-lineage law with a clean counterexample at (64,42). This one crosses the lineage.

Two consequences follow. First, both classical seed generators have one spectral shape: support exactly $\{0, 1, N-1, N/2\}$, with equal signs at $\{0, N/2\}$ and the opposite sign at $\{1, N-1\}$. The predecessor derives them by unrelated arguments — unit rank one with a fundamental unit at octal, unit rank zero with argument conditions doing the work at hexagonal — and never places the results side by side.

Second, inheritance and nativity separate by support rather than by dimension count. At (16,24) the inherited class occupies exactly the replicated seed support and nothing outside it; the two native classes occupy that set and six further bins. Read modulo the seed period against the orbit factorisation of $x^8 - 16$, the twenty-four bins partition three ways: $\{0, 1, 4, 7\}$ the unit orbits, inherited; $\{2, 6\}$ the 3-orbit, opened by subdivision; $\{3, 5\}$ the 5-orbit, untouched by any of the five directions of R . The reversal reported in the predecessor becomes a statement about which bins, available before any dimension is counted.

New measurement: (64,30), $\dim R = 1$, stable across a 100x change in coefficient bound and forty additional digits. It fills the empty cell in the predecessor's lineage table and retires the reading that the hexagonal dimension climbs monotonically; the sequence is 1, 2, 3, 4, 1, 1. The predecessor's open problem asks for a fifth point on $\gcd(q, 12)$ and gets one, but a second candidate fits the same six values and the discriminator lies at $q = 8$.

Two corrections to the predecessor. Its instrument taxonomy separates failure modes but not axes: coefficient bound and working precision are one axis, coefficient field is another, and they cure different

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github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

failures — at (16,12) a 10^4 -fold increase in coefficient bound changes nothing while one adjoined element resolves the relation at unchanged precision. And its closing pattern — every backward derivation survived — is contradicted by its own Section 4: the forcing lemma is a backward derivation that shipped with a missing hypothesis. Both of its routes were evaluated at (16,8) and (64,6), and at both frames no unshared prime divides N, so the missing hypothesis held vacuously. The operative notion is not method diversity. It is case-set spanning.

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1. What this paper adds, and what it does not

This is not a new predicate. The governing rule remains the one stated in the predecessor: given a distinction inside a structure, ask whether it has a partner it can cancel against; no partner forces, a partner frees. Everything below is that rule read in a different representation, and the paper is organised so that each section names the representation rather than renaming the rule.

What is new is the representation itself and four measurements taken in it:

- (1) The realisation map, run forward, is a discrete Fourier transform with integral output, and both classical seeds have the same spectral support and sign pattern.
- (2) Subdivision is zero insertion; the transform commutes with it and produces exact periodic replication. This is the mechanism behind the predecessor's deterministic lift heights, and unlike the two-factor gate it holds on both lineages.
- (3) Inherited and native structure separate by spectral support, and the separating bins are the orbits of the seed's own factorisation. Nativity becomes visible before a dimension is counted.
- (4) (64,30) is measured, filling the one empty cell in the predecessor's lineage table.

Two methodological corrections are reported in Sections 9 and 10, both of which amend statements the predecessor makes about itself.

2. Setup

Notation follows the predecessor exactly and is not restated in full. Fix an integer base b with $|b| > 1$ and a period N . The shell sums are $S_r = \sum_{k \geq 0} 1/((Nk+r) b^k)$ for $r = 1..N-1$, the $r = 0$ shell being divergent and excluded. K is the lattice of integer vectors c with $\sum c_r S_r = 0$, and $C(y) = \sum c_r y^r$, so $C(0) = 0$ by construction. Put $\theta = b^{1/N}$, $\omega = \zeta_N$, $\alpha_j = \theta \omega^j$, $\beta_j = 1 - 1/\alpha_j$, and $L_j = \log \beta_j$ on the principal branch. R is the rational relation lattice among the L_j , $\dim K$ over the algebraic closure equals $\dim R$ by Baker, and $\dim R$ is the frame-intrinsic quantity.

The single object this paper works with is the realisation map of the predecessor's Appendix A, which sends a relation vector w in R to a coefficient vector:

$$c_r = \lambda * w_{\hat{r}} / (N \theta^{\hat{r}}), \quad w_{\hat{r}} = \sum_j w_j \omega^{\alpha_j(-j \cdot r)}$$

The predecessor uses this map in one direction only, to realise a known w over a named field. Running it in the other direction is the whole of Sections 3 through 7.

3. The forward transform, and one shape for two seeds

Inverting the realisation map gives a forward transform on the coefficient vector. Writing $a_r = c_r \theta^r$,

$$w_j = (1/N) \sum_r (c_r \theta^{\hat{r}}) \omega^{\alpha_j(-j \cdot r)}$$

which is a discrete Fourier transform of the sequence a . The derivation is one line from the closed form. Since $\sum_j \zeta^{(-jr)} L_j = n \theta^{(-r)} S_r$, the shell relation $\sum_r c_r S_r = 0$ becomes $\sum_j w_j L_j = 0$ with w_j as above. So the coefficient vector and the relation vector are a transform pair, not two descriptions.

Run on both classical seeds at 120 digits:

seed	b	N	theta	$w = \text{DFT}(c_r \theta^{ra})$	$\max \text{Im } w_j $
octal	16	8	$\sqrt{2}$	$[-8, 8, 0, 0, -8, 0, 0, 8]$	$3.5e-120$
hexagonal	64	6	2	$[-32, 32, 0, -32, 0, 32]$	$6.4e-119$

Both outputs are integral, and both agree with the predecessor's independently derived generators up to scale: the octal vector is -8 times its Section 4.1 generator $(1, -1, 0, 0, 1, 0, 0, -1)$, and the hexagonal is 32 times its Section 4.2 generator $(-1, 1, 0, -1, 0, 1)$. That agreement is a consistency check, not a result. The result is what the two have in common.

Proposition 3.1 (seed shape). Both classical seed generators are supported exactly on $\{0, 1, N-1, N/2\}$, with equal signs at $\{0, N/2\}$ and the opposite sign at $\{1, N-1\}$.

seed	N	support	$= \{0, 1, N-1, N/2\}$	sign at $\{0, N/2\}$	sign at $\{1, N-1\}$
octal	8	$[0, 1, 4, 7]$	TRUE	+, +	-, -
hexagonal	6	$[0, 1, 3, 5]$	TRUE	-, -	+, +

The two are identical up to the global sign, which is the free scalar λ and carries no information (Section 5). This is worth recording because the predecessor obtains these two vectors by arguments that share nothing. At octal, $\theta = \sqrt{2}$ lies in $\mathbb{Q}(\zeta_8)$, the field is $\mathbb{Q}(\zeta_8)$ of degree 4 with unit rank one, and the fundamental unit $1 + \sqrt{2}$ does the separating work. At hexagonal, $\theta = 2$ is rational, the field is $\mathbb{Q}(\sqrt{-3})$ of degree 2 with unit rank zero, units are torsion only, and the argument conditions carry the argument — the predecessor notes explicitly that valuations alone leave three dimensions there and select nothing. Different fields, different unit ranks, different constraint families, and the same four occupied bins with the same sign pattern.

No mechanism is claimed for this. It is a measured coincidence across the only two known positive-base seeds, and Section 12 records what would settle it: the third seed at $(-64, 12)$, whose two directions have not been transformed.

4. Winding is zero insertion

The predecessor's Section 7.1 establishes that $W \cdot K_{\text{seed}}$ is contained in K_{frame} , strictly, and that lift heights multiply exactly: source height times cofactor height, $8 \times 16 = 128$ into $(256, 16)$, $8 \times 256 = 2048$ into $(4096, 24)$, $24 \times 64 = 1536$ into $(4096, 12)$. Height multiplication is the visible consequence of polynomial multiplication. The transform says what that multiplication is.

4.1 The index and the radical rescale together

Let (b, n) be a seed with coefficient vector c and relation vector w , and let (b, nq) be its subdivision. The lift places c at every q -th coordinate:

$$d_{\lfloor qr \rfloor} = c_r, \quad d_{\lfloor qr \rfloor + s} = 0 \quad \text{for } 0 < s < q$$

This is literally zero insertion by a factor q . Now transform the lifted vector. Its sequence is $A_m = d_m \theta_{\lfloor nq \rfloor}^m$, and at the occupied positions $m = qr$,

$$A_{\lfloor qr \rfloor} = c_r \theta_{\lfloor nq \rfloor}^{\lfloor qr \rfloor} = c_r (\theta_{\lfloor nq \rfloor}^q)^r = c_r \theta_n^r = a_r$$

because $\theta_{\lfloor nq \rfloor}^q = (b^{1/nq})^q = b^{1/n} = \theta_n$. The radical rescaling exactly compensates the index rescaling. That single identity is the whole mechanism, and it is why the transform commutes with the winding map rather than merely being compatible with it.

Theorem 4.1 (replication). With A as above, the nq -point transform of the lifted vector satisfies $W_k = (1/q) w_{(k \bmod n)}$ for every k . Hence the occupied bins of the lift are the union over $t = 0 \dots q-1$ of $\text{supp}(w) + tn$, and the amplitude is scaled by $1/q$ from the normalisation.

Proof. $W_k = (1/(nq)) \sum_m A_m \zeta_{\lfloor nq \rfloor}^{(-km)} = (1/(nq)) \sum_{r=0}^{n-1} a_r \zeta_{\lfloor nq \rfloor}^{(-kqr)} = (1/(nq)) \sum_r a_r \zeta_n^{(-kr)} = (n/(nq)) w_{(k \bmod n)}$.

4.2 Measured

Three tests, at increasing strength: a frame already known, an inverse, and a frame the law had not seen.

test	frame	q	predicted bins	measured bins	max deviation
replication	(16,24)	3	{0,1,4,7}+{0,8,16}	identical	7.6e-140
inverse	(16,24) -> (16,8)	3	seed recovered	[0,8,-8,-4,-8,-2,-2,1]	6.3e-140
forward, unseen	(16,16)	2	{0,1,4,7}+{0,8}	identical	3.2e-139

At (16,24) the measured amplitudes are $\pm 8/3$ at the twelve predicted bins and zero elsewhere, at 140 digits. The inverse collapses the three spectral blocks, undoes the $1/3$ gain, transforms back, and divides out θ^r , returning the seed vector [8, -8, -4, -8, -2, -2, 1] exactly. Forward and inverse both close, so subdivision loses nothing in the transform domain; the inclusion of the predecessor's Section 7.1 is strict in K but the map onto the inherited subspace is invertible.

At (16,16) the coefficient vector was obtained from scratch by iterated detection with no seed information supplied. The bins matched the prediction exactly. The amplitudes came back negated, because the detector returned $-c$; that is the free scalar λ and is treated in the next section.

4.3 The second lineage

The predecessor's two-factor gate is verified on five octal frames and fails one lineage over at (64,42); its Appendix D states it as an octal-lineage law for that reason. The same discipline is applied here before any general claim: the replication law was verified on three octal frames, all at $b = 16$, and the hexagonal lineage is the test that matters.

frame	seed	q	zero-inserted vector a relation?	bins match	max deviation	inverse
(64,18)	(64,6)	3	YES, residual 3.776×10^{-201}	TRUE	6.1×10^{-199}	3.658×10^{-199}

The hexagonal seed [16, -24, -8, -6, 1] zero-inserted by three is a genuine relation at (64,18) at residual 3.8×10^{-201} , its spectrum is the hexagonal seed's four bins replicated three times across eighteen at bins {0,1,3,5,6,7,9,11,12,13,15,17}, the deviation from $(1/3) w_{(k \bmod 6)}$ is 6.1×10^{-199} , and the inverse returns [0, 16, -24, -8, -6, 1] to 3.7×10^{-199} .

The law crosses the lineage that the gate does not. That is the difference between a statement about a particular partner structure and a statement about the representation, and it is the reason this section is stated as a theorem while Section 9 of the predecessor is stated as a measurement.

5. Support is gauge-invariant; amplitude is not

A relation is defined up to a nonzero scalar, and the realisation map carries that freedom explicitly as λ . The predecessor's instrument ledger records the cost of forgetting it: fixing $\lambda = 1$ in the interpolation reported two of five directions at (16,24) as unrealisable, and the fix is that the antisymmetric part needs $\lambda = i$.

The (16,16) test above supplies the positive statement. The detector returned the negated seed, so the measured amplitudes were the predicted amplitudes times -1, which registers as a deviation of 8.0 against a prediction of magnitude 4. Absorbing the scalar, the deviation is 3.2×10^{-139} .

quantity	status under $c \rightarrow \lambda c$	measured at (16,16)
occupied bin set	invariant	identical to prediction
amplitude	covariant, scales by λ	matched after absorbing $\lambda = -1$
deviation, λ ignored	meaningless	8.0
deviation, λ absorbed	the real quantity	3.2×10^{-139}

So the invariant content of a relation, in this representation, is its support. That is consistent with the predecessor's own finding that scalar readouts are reader-dependent while structural relations survive representation change, and it is the reason Sections 6 and 7 report bins rather than numbers.

6. Native structure is support excess

The predecessor computes nativity as a dimension: $\dim K$ at the frame minus the dimension of the ancestral lifts, giving native = 2 at (16,24) and, after the negative-base ancestors are restored, native = 1 at (4096,24). That is a subtraction of counts and it inherits every difficulty of counting. In the transform domain the same distinction is a set difference.

At (16,24) the grading modulus is $d = 3$ and equals the subdivision index, so by the predecessor's Corollary to Theorem 6.1 class-pure detection is complete. The three rational classes were detected separately and transformed:

class	c-support	height	residual	spectral support outside replicated seed set
$r = 0 \bmod 3$	3,6,9,12,15,18,21	8	1.75e-301	(empty)
$r = 1 \bmod 3$	1,4,7,...,22	32	2.39e-301	2, 6, 10, 14, 18, 22
$r = 2 \bmod 3$	2,5,8,...,23	48	5.48e-301	2, 6, 10, 14, 18, 22

Class 0 is the inherited lift. Its spectral support is exactly the replicated seed support $\{0,1,4,7,8,9,12,15,16,17,20,23\}$ with nothing outside it, which is Theorem 4.1 in the form of a measurement. The two native classes occupy that entire set and six further bins.

Proposition 6.1. Inherited structure occupies exactly the replicated seed support. Native structure occupies that set and more; the excess is the native signature.

Inheritance and nativity are therefore not spectrally disjoint. A working hypothesis stated earlier in the session that produced this paper — that the native directions sit in bins the replication does not reach — is false and is recorded in Section 11. They sit in those bins and in others, and the discriminator is the excess, not the location.

The practical consequence is that a kernel vector can be classified as inherited or native by transforming it and testing set containment, with no ancestor enumeration and no quotient. The predecessor's Section 7.2 records that its ancestor enumerator contained a guard rejecting roots below 2, which made the entire negative half of the family unreachable and inflated native from 1 to 3. A support test has no enumerator to put a guard in.

7. The three-way partition, and the reversal as a support fact

The six excess bins at (16,24) are $\{2, 6, 10, 14, 18, 22\}$. Reduced modulo the seed period 8 they are $\{2, 6\}$. The twelve inherited bins reduce to $\{0, 1, 4, 7\}$. Six bins are left over: $\{3, 5, 11, 13, 19, 21\}$, reducing to $\{3, 5\}$.

Those three sets are not arbitrary. They are the orbits of the seed's own factorisation, obtained from $x^8 - 16$ by exact factorisation with no numerical input:

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x^2 - 2*x + 2  g(1) = 1  primes {}   bins j = [1, 7]  unit
x^2 - 2        g(1) = -1 primes {}   bins j = [0, 4]  unit
x^2 + 2        g(1) = 3  primes {3}  bins j = [2, 6]  3-channel
x^2 + 2*x + 2  g(1) = 5  primes {5}  bins j = [3, 5]  5-channel

```

bins mod 8	orbit	g(1)	status at (16,24)
$\{0, 4\}$ and $\{1, 7\}$	unit orbits	-1, 1	inherited; the replicated seed support
$\{2, 6\}$	3-channel	3	native; opened by the subdivision
$\{3, 5\}$	5-channel	5	unoccupied by any direction of R

The third row required a separate measurement, because the absence of those bins from the three rational classes proves only that the rational reader does not reach them. R was therefore computed directly at

(16,24) by the predecessor's conjugation split, returning $\dim R^+ = 3$ and $\dim R^- = 2$, matching its Section 8 table. None of the five directions touches $\{3, 5, 11, 13, 19, 21\}$. Those bins are unoccupied at the ceiling, not merely unread.

This is the predecessor's Section 9.3 reversal, seen one level down. There the statement is that an unshared channel opens if and only if its prime divides the period, established by $\dim R = 5$ at (16,24) and (16,40) with the identical 3-symmetric, 2-antisymmetric signature and the channels exchanging roles. Here the same fact appears as which bins are occupied, and it is available from the support before any dimension is counted. A prediction and its confirmation that agree at the level of counts agree here at the level of addresses.

Recorded as a limitation: this partition is measured at one frame. The analogous decomposition at (16,40), where the roles exchange, has not been transformed, and Section 12 lists it as the first thing to run.

8. (64,30), and the hexagonal amendment

The predecessor's Section 10 tabulates the two lineages across subdivision index and leaves one cell empty — the hexagonal lineage at $q = 5$. Its Section 13 states the consequence precisely: four data points fit $\dim R = \gcd(q,12)$, and with five points that would be a fit rather than a law.

The frame is (64,30). Measured by conjugation split at two coefficient bounds and two precisions, following the predecessor's Appendix A stability requirement:

frame	q	maxcoeff	dps	sym	anti	dim R
(64,24)	4	1e6	220	2	2	4
(64,24)	4	1e8	260	2	2	4
(64,30)	5	1e6	220	1	0	1
(64,30)	5	1e8	260	1	0	1
(64,42)	7	1e6	220	1	0	1
(64,42)	7	1e8	260	1	0	1

The completed row is 1, 2, 3, 4, 1, 1 across $q = 1, 2, 3, 4, 5, 7$.

This retires the predecessor's reading of its own table. Section 10 there says the hexagonal dimension climbs monotonically with q through 1,2,3,4 and then drops to 1 at $q = 7$ where the shared prime's divisibility no longer supplies the increment. Neither clause survives. The sequence is not monotone, and divisibility by the shared prime 3 is not the increment, since 3 divides neither 2 nor 4 and both climb.

What the six points do support is a gate on the seed period: $\dim R = q$ when $\gcd(q, 6) > 1$ and $\dim R = 1$ otherwise. That accounts for all six including both drops, and it means the hexagonal lineage gates as well as counts — it gates on $\gcd(q, N_{\text{seed}})$ and counts only once the gate is open, where the octal lineage gates on $\gcd(q, b-1)$. Both cross-applications were tested and both fail: $\gcd(q,15)$ predicts hexagonal opening at $q = 7$, which it does not, and $\gcd(q,8)$ predicts octal opening at $q = 2$ and 4, which it does not.

The predecessor's $\gcd(q,12)$ also survives the fifth point, since $\gcd(5,12) = 1$. The two candidates are observationally identical on all six measurements and first differ at $q = 8$, where $\gcd(8,12) = 4$ while the

conditional rule gives 8. Section 12 records (64,48) as the discriminator, with the caveat that forty-seven shells is at the edge of where the conjugation-split reduction remains tractable.

This is a case where the predecessor's open-problem list named the required measurement before the measurement existed, the measurement was taken, and the honest outcome is that the fit survives with a rival. The line in Section 13 stands as written and is now better supported and better bounded.

9. Two axes, not one scale

The predecessor's instrument ledger separates failure modes and pairs each with a remedy — a broken probe needs a positive control, a wrong baseline a derived null, a narrow reader an independently computed ceiling, a wrong address a second route, a basis-dependent count the invariant computed directly. What it does not separate is the two axes along which an instrument can be inadequate, and they behave differently enough that the distinction is worth stating.

The frame is (16,12), the predecessor's holdout, which reads zero over the rationals and carries a relation over $Q(2^{1/3})$ at residual 5.1×10^{-260} . Detection was run at four precisions and two coefficient bounds:

dps	maxcoeff 1e6	maxcoeff 1e10	reading
40	11-coefficient vector	identical vector	blur: cured by digits
80	different 11-vector	identical vector	blur: cured by digits
160	none	none	alias: not curable on this axis
240	none	none	alias: not curable on this axis

The first two rows are the mantissa-exhaustion failure the predecessor's Appendix A already warns about. They are reproducible, stable under a 10^4 -fold change in coefficient bound, and they evaporate when digits are added. The last two rows are a different object. They are correct about the samples taken, stable across six orders of magnitude of additional effort on that axis, and internally flawless; nothing inside the rational reader indicates anything is wrong. Adding one element resolves it immediately at the same precision, and the relation appears with coefficients in Q on $r = 2 \bmod 3$, in θQ on $r = 1 \bmod 3$, and zero on $r = 0 \bmod 3$.

The two axes are therefore capacity — working precision and coefficient bound — and dimension — the coefficient field. Capacity cures the first failure and cannot touch the second; dimension cures the second at unchanged capacity. The dangerous combination is high capacity with low dimension, because it produces a result that is exact, repeatable, internally consistent and incomplete, and additional effort along the capacity axis confirms it more firmly.

A second measurement quantifies what the low-dimension reader is missing, in a form that is cheap to compute. Given two real values stated to five decimal places, the number of integer relations $a x + b y = 0$ consistent with that reading, over $|a| \leq 2000$ and with propagated uncertainty, is 50. At a resolution of $1e-7$ it is 1, and at $1e-9$ it is 0. That count is the size of the set of distinct structures sharing one address at a given resolution, and it is the honest version of what a reader has when it has a value.

10. Route independence is case-set spanning

The predecessor closes its ledger with a pattern: every forward claim died and every backward derivation survived. The claim is attractive and this paper's own results were obtained in the backward direction. It is nonetheless contradicted by the predecessor's Section 4, and the counterexample is the paper's central lemma.

The forcing lemma is a backward derivation. It shipped with a missing hypothesis — the requirement that p not divide N — and the predecessor records the discovery in its ledger. So a backward derivation failed. It also had two routes behind it, and they were genuinely different: a valuation argument, and independent numerical verification of the resulting seed kernels at residuals 4.9×10^{-80} and 4.1×10^{-80} . Two routes agreed and the lemma was still incomplete.

The reason is visible in one table:

frame	b-1	unshared primes	any p dividing N?	role
(16,8)	$3 * 5$	3, 5	none	verification frame
(64,6)	$3^2 * 7$	7	none	verification frame
(16,12)	$3 * 5$	3, 5	3	later counterexample
(4,6)	3	3	3	later counterexample
(16,24)	$3 * 5$	3, 5	3	gate frame
(16,40)	$3 * 5$	3, 5	5	gate frame

Both verification frames satisfy the missing hypothesis vacuously. Neither route could detect that the hypothesis was doing work, because at neither frame was it doing any. Different methods evaluated on inputs that do not vary the hypothesis at issue are, for the purpose of testing that hypothesis, one route.

The operative notion is therefore not direction and not method diversity. It is whether the case set spans the hypothesis. This is procedural in a way that 'reason backward' is not: before promoting a claim, name the hypothesis and check that the test frames actually vary it.

Applied to the claims in this paper. Theorem 4.1 is verified at $q = 2$ and $q = 3$, on two bases, on two lineages, forward and inverse; the hypothesis that varies is the lineage, and it varies. The $\gcd(q,6)$ reading of Section 8 rests on six points all at $b = 64$ and all with $q < 8$, and the rival candidate first differs at $q = 8$, so the case set does not span the question and Section 8 says so. Proposition 3.1 rests on two seeds and is stated as a coincidence for exactly that reason.

Applied to the predecessor. Its two-factor gate is verified on five frames all at $b = 16$; the hypothesis 'the frame is a subdivision of a seed' does not vary within that set, and the boundary was found only by moving to (64,42) in another lineage — which the predecessor did, and reported, and used to restrict the claim. The practice was already correct. The summary paragraph is what needs amending.

11. Falsification ledger

11.1 Predecessor claims amended by measurement

claim	status	evidence
hexagonal dim R climbs monotonically with q	false	(64,30): 1,2,3,4,1,1 across q = 1,2,3,4,5,7
the increment is the shared prime's divisibility	false	3 divides neither 2 nor 4; both climb
every backward derivation survived	false	forcing lemma shipped without p not dividing N
instrument taxonomy is complete as stated	incomplete	modes are separated; the two axes are not

11.2 Claims originated and killed inside this work

claim	how it died
native directions sit in bins replication does not reach	they occupy the replicated bins and six more; the excess is the signature
the untouched bins {3,5} mod 8 hold the sqrt3 directions	direct computation of R at (16,24): none of the five directions touches them
dim R = q mod 6 on the hexagonal lineage	fit four points, killed at (64,30) on the first forward test
a false-negative check is two coefficient bounds	two bounds on one instrument is one route; the split is the second route

11.3 Instrument failures in this work

failure	effect	fix
PSLQ tolerance set one notch tighter than the coarsening error	a false focal plane: the true relation appeared at 30 digits and vanished at 50 and 90	tolerance must exceed (sum of c) times the coarsening error
class-wise detection at maxcoeff 1e7 at (16,40)	reported no relation where dim R = 5	compute dim R by conjugation split, a different object, not a larger bound
global sign not quotiented before comparing amplitudes	deviation reported as 8.0 where it is 3.2e-139	absorb lambda; compare supports

All three are the predecessor's own catalogued modes recurring in new code: a wrong baseline, a narrow reader, and a basis-dependent comparison. The second is the same class as its maxcoeff 10⁶ false negative at (16777216,24). That an instrument taxonomy derived from one session's failures immediately catches the next session's failures is the strongest evidence available that the taxonomy is real.

12. Open problems

THE OCCUPANCY MECHANISM, unchanged. Freeing is derived; what supplies weight to a freed coordinate is not. The transform representation does not close this, but it changes what a solution must produce: an account of which bins acquire support, not how many dimensions appear.

THE PARTITION AT (16,40). Section 7 measures the three-way partition at one frame. At (16,40) the channels exchange roles, so the prediction is that $\{3,5\} \bmod 8$ becomes the native excess and $\{2,6\} \bmod 8$ becomes unoccupied at the ceiling — an exact reversal in the support, testable by the same procedure.

THE THIRD SEED'S SPECTRUM. Proposition 3.1 rests on two seeds. The frame (-64,12) carries dimension 2 and both directions are native to it. Whether either has the shape $\{0, 1, N-1, N/2\}$ would move that proposition from coincidence toward structure, or kill it. The negative base requires the half-step offset in the root layout and the predecessor's Appendix B.3 already implements it.

(64,48). The discriminator between $\dim R = \gcd(q,12)$ and the conditional $\gcd(q,6)$ rule. Forty-seven shells is at the edge of tractability for the conjugation split; the predecessor's Appendix A records that lattice reduction did not terminate at comparable size.

REPLICATION AT COMPOSITE q AND UNDER WINDING. Theorem 4.1 is proved for subdivision. Winding raises base and period together, and the predecessor's Section 6.1 records that conflating the two inverted several readings. The transform-domain statement of winding, as opposed to subdivision, has not been derived.

SUPPORT AND OCCUPANCY IN THE HEXAGONAL LINEAGE. The predecessor reports that hexagonal bins $\{0,1,3,5\}$ are occupied at every q tested and the 7-channel never opens. Whether the inherited support there is likewise exactly the replicated seed support, and whether anything ever appears outside it, would say whether the accumulator behaviour is support growth or amplitude growth. Section 4.3 establishes only that the inherited part replicates.

Appendix A. Methods

Shell evaluation. Direct summation to $\text{ceil}(3.5 \text{ dps} / \log_{10}|b|) + 120$ terms at dps 140 to 300, following the predecessor.

The transform. Computed directly as $w_j = (1/N) \sum_r c_r \theta^r \omega^{(-jr)}$ in mpmath at dps 120 to 260, with $\omega = \exp(2\pi i / N)$ and θ the real positive root for $b > 0$. Integrality of the output is verified by detection against the rational power basis; the imaginary parts are reported as a floor check and run at 10^{-119} or below in every case.

Support. A bin is occupied when $|w_j|$ exceeds 10^{-35} to 10^{-60} depending on the working precision, always at least 10^{-60} above the observed floor for that computation. No bin in this work fell within three orders of magnitude of the threshold.

The relation lattice R. Conjugation split exactly as in the predecessor's Appendix A: symmetric part from $\text{Re } L$ on $j = 0..N/2$, antisymmetric from $\text{Im } L$ on $j = 1..N/2-1$, each by iterated detection with elimination of the largest coordinate. Residual gate at $10^{-(0.75 \text{ dps})}$, detection tolerance at $10^{-(0.82 \text{ dps})}$.

Stability. Every dimension reported in Section 8 was computed at maxcoeff 10^6 and 10^8 and at dps 220 and 260. The (16,12) sweep of Section 9 was run at maxcoeff 10^6 and 10^{10} . No dimension in this paper is reported from a single instrument setting.

Appendix B. Code

B.1 The forward transform and the seed shape.

```
import mpmath as mp
mp.mp.dps = 120

def dft(c, b, N):
    th = mp.mpf(b)*(mp.mpf(1)/N); z = mp.exp(2j*mp.pi/N)
    a = [mp.mpf(c[r])*th**r for r in range(N)]
    return [sum(a[r]*z**(-j*r) for r in range(N))/N for j in range(N)]

dft([0, 8, -8, -4, -8, -2, -2, 1], 16, 8) # -> [-8, 8, 0, 0, -8, 0, 0, 8]
dft([0, 16, -24, -8, -6, 1], 64, 6) # -> [-32, 32, 0, -32, 0, 32]
# support {0, 1, N-1, N/2} in both; signs equal at {0, N/2}, opposite at {1, N-1}
```

B.2 Replication, its inverse, and the cross-lineage test.

```
SEED = [0, 8, -8, -4, -8, -2, -2, 1] # octal, N = 8
w8 = dft(SEED, 16, 8)

# subdivision by q = 3: zero insertion
d24 = [0]*24
for r in range(8): d24[3*r] = SEED[r]
W24 = dft(d24, 16, 24)
```

```

assert max(abs(W24[k] - w8[k % 8]/3) for k in range(24)) < 1e-139

# inverse: collapse the q blocks, undo the gain, transform back, divide out theta^r
w_rec = [3*W24[j] for j in range(8)]
th8, z8 = mp.sqrt(2), mp.exp(2j*mp.pi/8)
a_rec = [sum(w_rec[j]*z8**(j*r) for j in range(8)) for r in range(8)]
c_rec = [mp.re(a_rec[r])/th8**r for r in range(8)] # -> SEED, error 6.3e-140

# hexagonal lineage, the test that the two-factor gate fails
HEX = [0, 16, -24, -8, -6, 1]; w6 = dft(HEX, 64, 6)
d18 = [0]*18
for r in range(6): d18[3*r] = HEX[r]
W18 = dft(d18, 64, 18)
assert max(abs(W18[k] - w6[k % 6]/3) for k in range(18)) < 1e-198

```

B.3 Native as support excess, and the orbit partition.

```

import sympy as sp
x = sp.Symbol('x')

# the three sets, from the seed's own factorisation -- no numerics
for g, _ in sp.factor_list(x**8 - 16)[1]:
    gp = sp.Poly(g, x)
    print(g, 'g(1) =', int(gp.eval(1)), sp.factorint(abs(int(gp.eval(1)))))
# x^2-2x+2 -> 1 {} bins {1,7} unit
# x^2-2 -> -1 {} bins {0,4} unit
# x^2+2 -> 3 {3:1} bins {2,6} 3-channel <- native excess at (16,24)
# x^2+2x+2 -> 5 {5:1} bins {3,5} 5-channel <- unoccupied at the ceiling

seed_supp = sorted({(j + 8*t) for j in [0,1,4,7] for t in range(3)})
def classify(c24):
    s = [j for j, v in enumerate(dft(c24, 16, 24)) if abs(v) > mp.mpf(10)**-35]
    return {'inherited': [j for j in s if j in seed_supp],
            'excess': [j for j in s if j not in seed_supp]}
# class 0 -> excess [] ; classes 1,2 -> excess [2,6,10,14,18,22]

```

B.4 (64,30). Conjugation split at two bounds, as required by Appendix A.

```

def dimR(b, N, maxc, dps):
    mp.mp.dps = dps
    th = mp.mpf(abs(b))**(mp.mpf(1)/N); w = mp.exp(2j*mp.pi/N)
    L = [mp.log(1 - 1/(th*w**j)) for j in range(N)]
    def rs(vals):

```

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github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

```

live, out = list(range(len(vals))), []
gate = mp.mpf(10)**(-int(dps*0.75))
while len(live) > 1:
    sub = [vals[i] for i in live]
    rel = mp.pslq(sub, maxcoeff=maxc, maxsteps=400000,
        tol=mp.mpf(10)**(-int(dps*0.82)))
    if rel is None: break
    if abs(mp.fsum(mp.mpf(c)*v for c, v in zip(rel, sub))) > gate: break
    out.append(rel)
    live.pop(max(range(len(rel)), key=lambda k: abs(rel[k])))
return out
sym = rs([mp.re(L[j]) for j in range(0, N//2 + 1)])
anti = rs([mp.im(L[j]) for j in range(1, N//2)])
return len(sym), len(anti)

dimR(64, 30, 10**6, 220) # (1, 0) -> dim R = 1
dimR(64, 30, 10**8, 260) # (1, 0) -> dim R = 1

```

Appendix C. Frames measured in this work

frame	q	what was measured	result
(16,8)	-	forward transform of the seed kernel	w = [-8,8,0,0,-8,0,0,8], support {0,1,4,7}
(64,6)	-	forward transform of the seed kernel	w = [-32,32,0,-32,0,32], support {0,1,3,5}
(16,16)	2	replication, forward prediction	bins exact; amplitude exact after lambda
(16,24)	3	replication, inverse, class supports	7.6e-140; 6.3e-140; excess {2,6} mod 8
(16,24)	3	dim R by conjugation split	3 sym + 2 anti = 5; {3,5} mod 8 untouched
(64,18)	3	replication, cross-lineage	6.1e-199; inverse 3.658e-199
(16,12)	-	capacity sweep vs field widening	blur at dps 40-80; alias at 160-240
(64,24)	4	dim R, two bounds	4 (2 sym, 2 anti), stable
(64,30)	5	dim R, two bounds	1 (1 sym, 0 anti), stable - NEW
(64,42)	7	dim R, two bounds	1 (1 sym, 0 anti), stable

Appendix D. Scope

Theorem 4.1 is proved for subdivision of a seed frame by an integer factor q, with the lift as defined in the predecessor's Section 7.2. It is verified at q = 2 and q = 3, on b = 16 and b = 64, forward and inverse. It is not verified at composite q, and no statement is made about winding, which raises base and period together.

Proposition 3.1 is a measured coincidence on the two known positive-base seeds. No mechanism is offered and none should be inferred. The third seed is the available test and it has not been run.

Section 6 measures the inherited-native support separation at one frame. Section 7's three-way partition is measured at that same frame, with the ceiling-level claim resting on a direct computation of R there. The reversal frame (16,40) has not been transformed.

Section 8's $\gcd(q,6)$ reading rests on six hexagonal measurements, all at $b = 64$, all at $q < 8$, and is stated alongside the rival it cannot yet be separated from. It is a fit.

Sections 9 and 10 are methodological. They report measurements and they amend statements the predecessor makes about its own practice. They are not mathematical results and nothing in Sections 3 through 8 depends on them.

The framework vocabulary that shaped this work — reader against object, absence as an operator, value against shape — appears nowhere in the derivations. Theorem 4.1 follows from $\theta_{(nq)}^q = \theta_n$ and the definition of the transform, and would be true if the framework were abandoned entirely.

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